

Before Down

2nd Presentation Report

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March 9 2023

1 Introduction

For this first presentation, Groupe I will detail what has been carried out since submitting the book of specifications. The group will detail what has been achieved ahead of or behind schedule and by whom and present what needs to be done for the next time.

After many code issues and hours on the project, Groupe I presents Before Dawn!



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2 Before Dawn

2.1 Reminder on what is Before Dawn

”Before Dawn” is a tower defense game. It is a real-time strategy video game played by building turrets or similar buildings on the map to stop the game’s enemies from reaching the fortress. Generally, enemies do not actively attack turrets. The player receives bonuses or points to buy turrets when an enemy is destroyed. Enemies attack in waves, and most tower defenses will pause after a wave to allow players to add turrets for points. Players generally have several hearts, and if a turret fails to destroy an enemy, the enemy will have fewer hearts once they reach a given place. As players upgrade their turrets, the number and ability of monsters will also increase. Generally speaking, monsters improve faster than turrets. The game’s objective is to survive for a certain amount of time or as long as possible.

2.2 Some new ideas

After many discussions and debates, we have decided to change and add some game mechanics; BeforeDawn became an exciting and thrilling game that will keep you on the edge of your seat. The game is reminiscent of the popular games Clash Royale and WarCraft3, but with a twist. The game has become different; it is no longer a simple defensive computer attack; the game now requires the player to train his troops and destroy the enemy’s main base through a robust military force to win. The whole game gets harder but also more exciting.

Instead of battling your opponent with troops and spells, you must mine gold and build defenses to protect your base. The game revolves around mining gold and using it to build defenses. You start with a limited amount of gold, but as you progress, you can mine more gold and build more elaborate defenses. You can construct walls, turrets, traps, and other defenses that will help protect your base from enemy attacks. The enemy player is doing the same thing, mining gold and building defenses to protect their base. As the game progresses, you must balance the need to mine gold with the need to build defenses. You must also keep an eye on your enemy’s base and try to find weaknesses in their defenses. Fewer hearts once they reach a given place. As players upgrade their turrets, the number and ability of monsters will also increase. Generally speaking, monsters improve faster than turrets. The game’s objective is to survive for a certain amount of time or as long as possible. This type of game has also another old name : RTS games.

3 What is the new idea exactly

3.1 General Description

As we said in the last chapter, this is an RTS game where the game aims to destroy the main base of the opposing enemy. Each player has their own territory.

Since enemy home bases do not explode on their own, we need to train some soldiers to attack and destroy the enemy home base. Soldiers do not grow out of the ground by themselves, so we need to build our own training camps to train our soldiers. Of course, training soldiers isn't free, it costs money, so we need to spend in-game 'gold' to train soldiers, and gold doesn't come from magic; we need to dig it up in the gold mines. Once the gold is mined, we can do many things, train soldiers, and build defense towers to protect our base.

The game is as simple as that and very easy to understand.

3.2 More details

3.2.1 Heroes

In the game, we will also include a hero, a character that the player can control. This hero can have his own skills, such as fireball, which generates much damage at once, and poison gas, which slows down the enemy.

3.2.2 Miners

The gold coins in the gold mine will not pop out on their own; we need to train miners to dig the gold out. Miners only have a specific HP which means that they will not be able to attack.

3.2.3 Troops

There are different types of soldiers: archers, knights ... There is a limited number of soldiers. The soldiers can fight each other. Soldiers cannot be controlled.

3.2.4 Training Camps

There are different types of training camps: archers, knights... There is a limited number of training camps. Training camps are destructible. Training camps form a group of soldiers. The camps are destructible.

3.2.5 Mine

There is only one mine in the field. There is a limited number of gold coins in the mine. Mines are destructible.

3.2.6 Tower Defense

The defensive towers are destructible. The number of defensive towers is limited. There are different types of defense towers.

3.2.7 Game start

In this game, the player starts with a set number of gold coins, a few miners, a certain number of soldiers, and a chosen hero. Additionally, the player also starts with a set number of defense towers to protect their resources and fend off enemy attacks.

Our game mechanics are very similar to Warcraft III and Clash Royale.



Figure 1: Gameplay example from Clash Royale



Figure 2: Gameplay example from Warcraft3

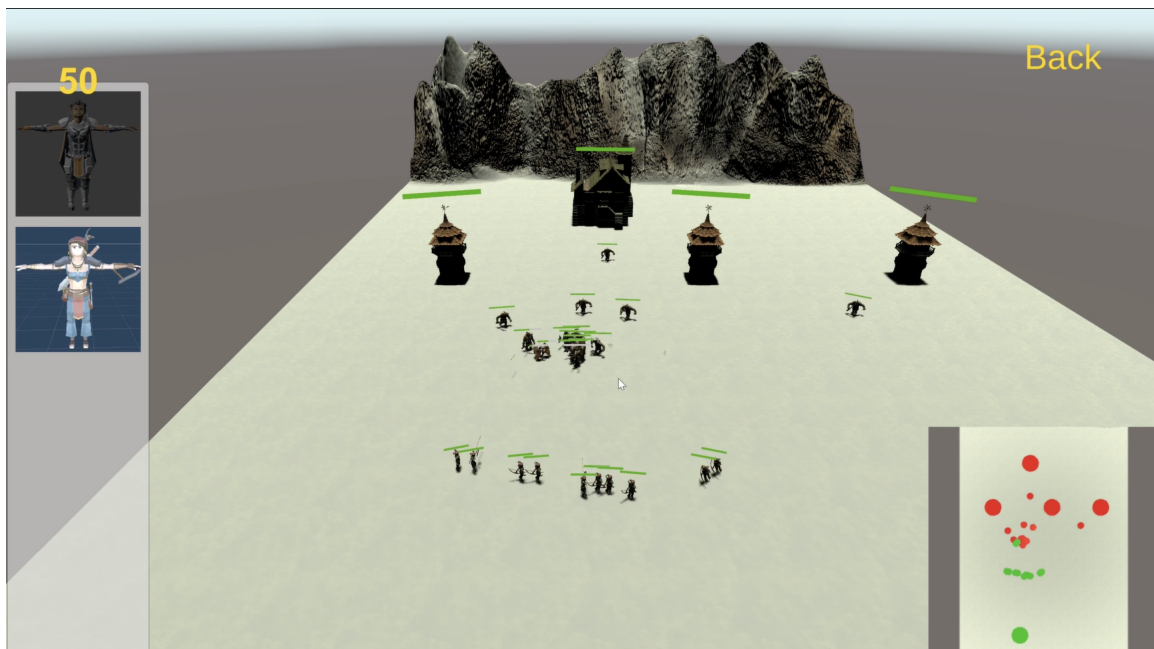


Figure 3: Gameplay example for what our team is looking for

4 What have been done

For this presentation, we focused on using unity and the programming part of the game's logic.

4.1 Unity

We did much work on game creation, especially learning Unity. Learning Unity was the first challenge for our group, as most of us in the group needed to gain experience with Unity. Almost half of the time we spent learning to use Unity.

4.2 Gameplay

The gameplay will progress simultaneously with the implementation of Artificial Intelligence and the User Interface in the game. For now, we have the map, which is not entirely finished; we use the Tilemap tools to make this map in 3D with free models taken from the internet: "Kenney."

4.2.1 Map

As stated in section 3.1, the entire map will be divided into two sections, with a river as the dividing line between two distinct but similar shapes. Our inspiration for this design was drawn from the popular game Clash Royale. The map will be composed of small squares, which is why the river is depicted as having a grid-like appearance rather than an irregular shape.

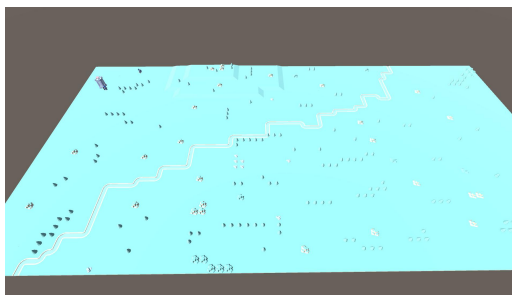


Figure 4: Map

Not only does this produce a beauty that gives the whole game its own originality, but it also makes it easier for us to make the script for the building of defense towers. We were also inspired by Warcraft 3 for the grid map.

4.2.2 PlayerClickingController

Moreover, We implemented a PlayerClickingController to let the player be able to control his Hero on the map by right-clicking; Warcraft III inspires this. The player-controlled character will not go into the tower or some trees using the mesh collider, and players can not go out. This makes the whole game much more logical.

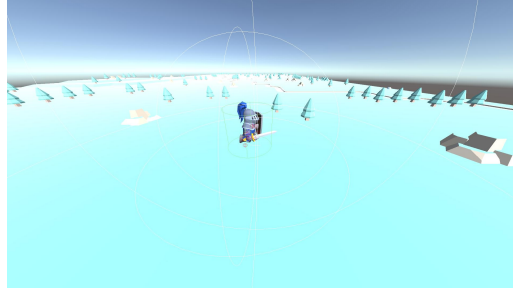


Figure 5: Character Controller

4.2.3 CameraController

As the map was quite large and we needed to allow the player to see the character models as well as facilitate the placement of defense towers, we needed to find a way to allow the player to manipulate the game camera.

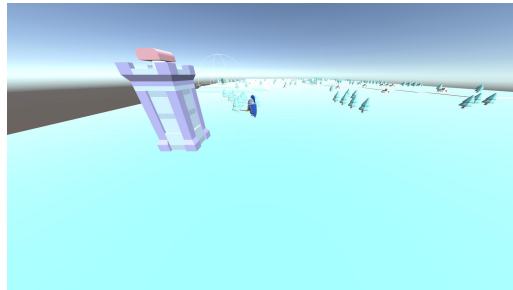


Figure 6: Camera Controller.

We also implemented the camera movement to let the player see the map. The camera is an RTS camera, players can move and zoom their mouse to see what is going on the map. Furthermore, we allowed the player to be able to place towers on the map. When the player wants to place his tower, we added a little detail that can make the gameplay more pleasant: when the player puts his mouse on the map, the color of the ground changes to red.

4.3 Artificial Intelligence

As we can see in 3.1, players need to create defense towers to defend themselves against enemy incursions, and the part about automatic defense tower attacks is essential.

4.3.1 Tower

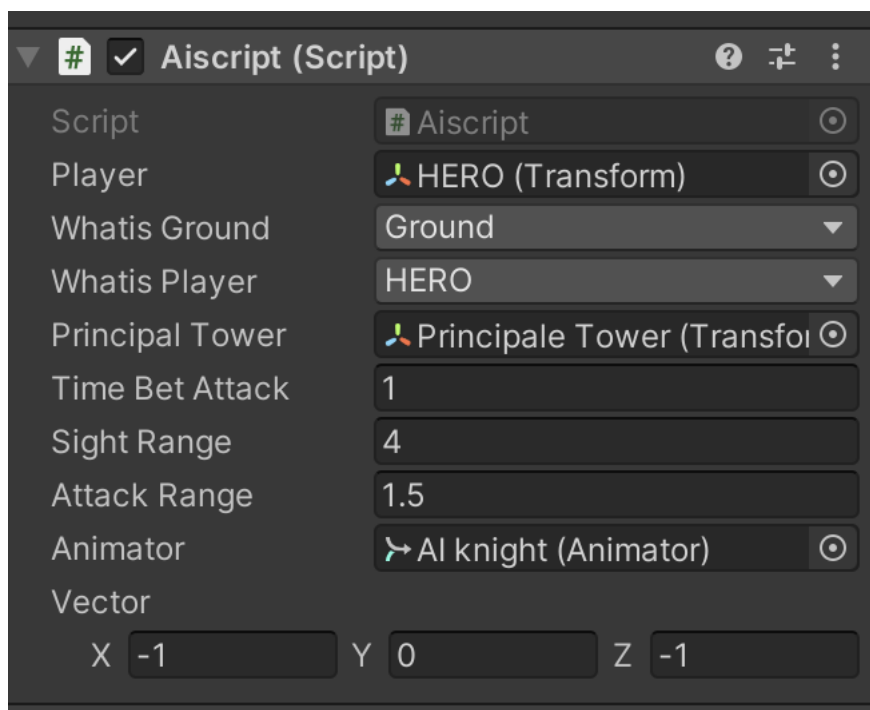
The progression on the AI part can be seen on the turret AI part, where the turret can follow one of the opponent's movements and shoot at it. Besides modular turrets, those settings can be easily changed to adapt to different towers.



Figure 7: Tower's AI.

4.3.2 Units

Moreover, Unit pattern AI will make the opponents target the central tower (Enemy base), but if those units meet turrets or their foes, they will attack them if they are within the range of attraction. Furthermore, they will disappear when their health point reaches 0. And last but not least, with hero AI, the player will be able to move the hero where he wants, and AI will find the shortest path on its own. We use many attributes to make the AI script. :



4

Figure 8: Attributes of AI Troops script.

4.4 Graphics

Graphics are a significant part of the game, and what makes players enjoy a game is the exciting gameplay and how good the models in the game look.

4.4.1 Problem

The graphics were more complicated than we expected; after three days with more than fifteen hours of work in Blender, we dropped our project but kept it if we continued later on; why this was so complex because All the project was done by hand with a computer mouse because none of us had a tablet.



Figure 9: This is an example made by ourselves.

4.4.2 Solution

That is why we found a solution to solve this problem: We need to find models on the internet!

Thanks to Mixamo, and Unity Free Assets, we found many different models of the characters on the internet and animations.

4.5 Sounds

The sound part of the game is also essential. An unforgettable game is not only a game with fascinating gameplay and incredible animations but also a game with sound effects that the player cannot forget after hearing it once. Many great games are reminded of it just by listening to the soundtrack.

4.5.1 Good news

We have found several good sound effects, which we have divided into three main categories: interactive button feedback, visual effects, and ambient effects.

Thanks to "Soundcrate," we find many sound effects.

Interactive button feedback sounds: whenever an in-game UI button is pressed, the game will give the player a sound back, such as the start game button, the pause game button, and the button to build a tower.

The visual performance sounds: we give the player a sound when the player attacks an enemy; we also give the player a sound when a defense tower attacks, the sound of a soldier attacking, the sound of a soldier dying, the sound of the player walking.

Ambient sounds: there is a soundtrack at the start of the game and a different soundtrack after the game has started.

4.5.2 Bad news

Because of the game's development schedule, we have only found these sound effects, but many still need to be added for use.

4.6 Gacha

Gacha is a random draw mechanic where a random selection is made to obtain a desired game item. It is widely used in many games, such as the famous Genshin Impact, which uses the gacha mechanic. Players who want to get the character they want need to get it by drawing. We also use the gacha mechanic in BeforeDawn, where the player must draw for a hero. The algorithm behind this system is pretty simple; we make sure that the drop rate of a hero is ten percent then we check if the player already has this hero to give him another reward, as mentioned after. A pity system also guarantees to drop a hero after ten draws. Each draw requires 100 game coins. If a hero is obtained that the player has already obtained, it will return 120 coins to the player.

[illegible]

Figure 10: This is Ganfaul.

4.7 Multiplayer

The preliminary phase of implementing the multiplayer aspect of our project has been completed successfully. We have established the necessary connectivity to enable multiple players to join the same server and participate in the game using a designated room name. Looking ahead to the next presentation, we plan to focus on developing scenes and the accompanying matchmaking settings, complete with UI menus. Specifically, we aim to address the placement of the two players on the game map and work on scripts that will govern the camera and point-of-view for each player.

We are confident that these developments will significantly enhance the user experience and improve the overall functionality of our project.

4.8 User Interface

The UI Part comes as one of the most exciting parts of the game, where we should create all the things the user can interact with, it was indeed complicated at some places to create it, but we managed to create the stuff needed for this presentation, and looking forward to creating more interesting UI stuff for the next presentation, especially the ones inside the game, as well as working on the designs and the good-looking of these UI's we created

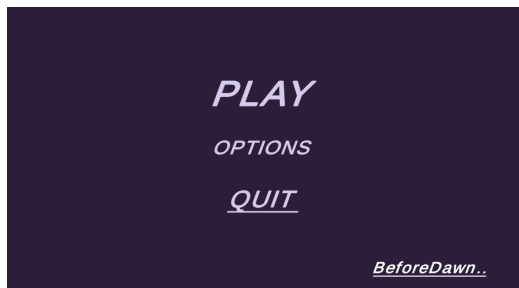


Figure 11: Main Menu

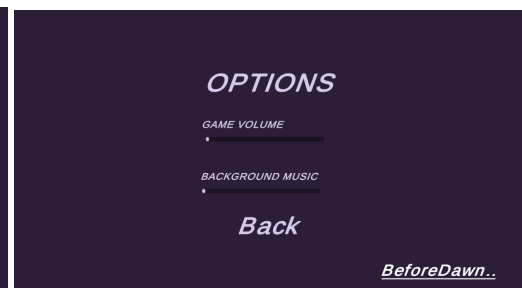


Figure 12: Options Menu

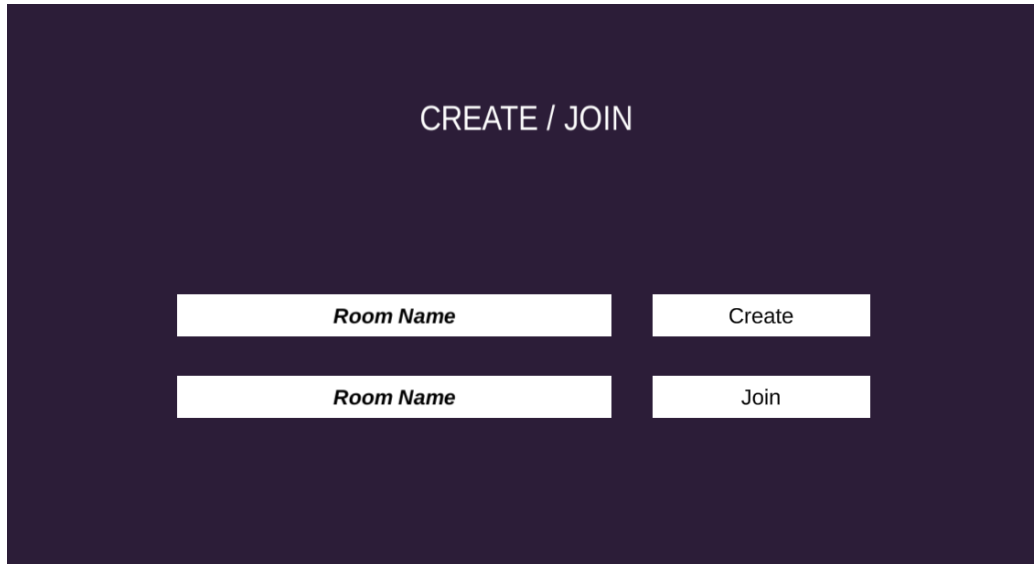


Figure 13: Create And Join Rooms

5 What is left to do

5.1 General description

So during the first presentation, approximately 30-35 percent of both AI (tower/Hero and Enemy) has been done, 20 percent of the gameplay has been completed, 40 percent of the Gacha task has been completed, and 30-35 percent of both UI tasks have been done.

After the first presentation, the main tasks it should progress on are the multiplayer and sound tasks. The multiplayer task should be done before the next presentation, so 100 percent, and for the sound, it must be caught up until the next presentation. The graphic task also needs to be caught up, but a solution has been found for this task.

A new task will also need to be implemented: it is the artificial intelligence for the miner. For the rest of the tasks, we will progress as indicated in the table below this part.

5.2 Gameplay

This part will follow the progressions of AI and UI. The bar life of each character in the game will be implemented, the victory or defeat at the end of the game must also be implemented, and a money system will start to be implemented. Moreover, towers and troops placement must also be completed before the next presentation, with a bar containing all types of turrets and troops. So before the next presentation, the game will be playable, but it will be not completely finished.



Figure 14: Health Bar

5.3 Graphics

As it was written, this task must be caught up, and the solution is to take many assets from the internet to make the game more beautiful and pleasant to our eyes.

5.4 Multiplayer

As it was written, this part must be finished before the next presentation. The multiplayer tasks will be developed using a unity asset: Photon. Thus the game will be playable for two people.

5.5 UI

The menu before the game and in the game will be partially complete until the next presentation. The player must be able to access the menu, change some parameters, and return to the game.

5.6 Sounds

The sound will be taken from the internet, the game's background sound will be ready and the sound for each action of the character will be implemented step by step.

5.7 Gacha

This task will be partially complete; all the code will be finished before the next presentation.

5.8 AI-(Unit Pattern)

In this task, Artificial Intelligence will be implemented for all types of troops. They can fight each other, attacking and dying from the enemy, even against Hero and tower.

5.9 AI-(Tower Pattern)

This task will also be implemented but must be completed. All types of turrets will be able to shoot, be destroyed, and take damage.

5.10 AI-(Miner Pattern)

This task should be completed before the next presentation. It will be able to spawn some miners to go to the mine and accumulate gold.

6 New organisation

The organization will change for some tasks:

- Gameplay : Florent will be in charge and Kun will assist
- Graphics : Florent will be in charge and Frederic will assist
- UI : Kun will be in charge and Daniel will assist
- AI-(Unit Pattern) : Florent will be in charge and Frederic will assist
- AI-(Tower Pattern) : Frederic will be in charge and Daniel will assist
- AI-(Miner Pattern) : Daniel will be in charge and Kun will assist

6.1 Distribution

Tasks / Name	Daniel	Florent	Frederic	Kun
Gameplay		In charge		Substitute
Graphics		In charge	Substitute	
Multiplayer	In charge	Substitute		
UI	Substitute			In charge
Sounds		Substitute		In charge
Gatcha	Substitute		In charge	
AI-(U.P)		In charge	Substitute	
AI-(T.P)	Substitute		In charge	
AI-(M.P)	In charge			Substitute

U.P = Unit Pattern

T.P = Tower Pattern

M.P = Miner Pattern

 In charge

 Substitute

6.2 Progression

Tasks / Time	First presentation	Second presentation	Final presentation
Gameplay	20%	60%	100%
Graphics	10%	60%	100%
Multiplayer	20%	99%	100%
UI	20%	50%	100%
Sounds	0%	50%	100%
Gatcha	40%	80%	100%
AI-(U.P)	30%	60%	100%
AI-(T.P)	30%	80%	100%
AI-(M.P)	0%	80%	100%

U.P = Unit Pattern

T.P = Tower Pattern

M.P = Miner Pattern

Many things have changed: First, the gameplay progress will be the same; we merge (UI-Game) and (UI-Menu) to UI, but the percentage progress will remain the same.

Nevertheless, the multiplayer, sound, and graphics are a lesser percentage than was for the first presentation. Except for the multiplayer, we have stated that these parts are less significant, and then we focused on the other parts, like AI. But we know we need the multiplayer option before the next presentation.

Finally, the percentage of both AI's have higher percentages than was for the first presentation, and these percentages will be even higher for the next one.

7 Conclusion

After an extended period of effortful collaboration, Groupe I is pleased to present the completed portion of our project. Despite several technical hurdles and compatibility concerns among team members, we have accomplished the assigned tasks. Moving forward, we anticipate delivering even more meaningful content for the subsequent presentation, such as new designs for our UI, multiplayer tasks completed, character introductions, related soundtracks, graphics refinement, game-play enhancements, and various other project improvements. In the following presentation, we look forward to showcasing our team's diligent work and commitment to project excellence.

Groupe I